

Prevalence and Pattern of Ocular Manifestations of HIV/AIDS among Adult Patients on Highly Active Anti-Retroviral Therapy in Federal Medical Centre Lokoja, Nigeria

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ABSTRACT

Reliable data on the magnitude and types of ocular manifestations of HIV/AIDS in a HIV-endemic setting is necessary in the current era of Highly Active Anti-Retroviral Therapy (HAART). Nigeria remains a high HIV burden country in Sub-Saharan Africa, and the young adults in Nigeria constitute a good percentage of all new HIV infections yearly. The nature and prevalence of ophthalmic problems in HIV-infected patients in developing countries differ from those in developed countries. Aim: This study aimed to determine the pattern and prevalence of ocular manifestations of HIV/AIDS among adult patients on HAART that presented at Federal Medical Centre, Lokoja, Kogi State. The study was a hospital-based cross-sectional study of HIV-positive patients on HAART at Federal Medical Centre, Lokoja, Kogi State. Patients' recruitment was by systematic random sampling. The study was done using pretest questionnaires administered by the researchers. Data was analysed using Statistical Package for Social Sciences (SPSS) version 21. A total of 310 participants, comprising 61 (19.7%) males and 249 (80.3%) females, were interviewed and examined. The male-to-female ratio was 1: 3.1. The age range was 21 years to 72 years. The commonest ocular manifestation was anterior uveitis (25%), while the least was cytomegalovirus retinitis (6.3%). Most lesions (37.5%) were found in the posterior segment. Prevalence of ocular manifestation in our hospital was low. Anterior uveitis and immune recovery uveitis are the most common ocular manifestations of HIV/AIDS that were seen among the study participants. The importance of ocular examination in HIV/AIDS-positive patients is paramount.

Keywords: Ocular manifestation, HIV/AIDS, prevalence, Lokoja, Nigeria.

INTRODUCTION

Acquired immunodeficiency syndrome (AIDS) is a group of health problems characterized by severe reduction or loss of the body's cellular immunity, which adversely affects the resistance to infection and malignancy.¹⁻³ It is caused by acquired immunodeficiency virus (HIV). AIDS was first described clinically in 1981 when five cases of pneumocystis carinii causing pneumocystis pneumonia were reported in previously healthy young men, and since then it has been a

pandemic especially in sub-Saharan Africa.⁴⁻⁶

The 2018 prevalence of HIV in Nigeria as released by the Federal Ministry of Health was 1.4%⁷, and the earlier years had the following: 4.1% in 2010, 3.4% in 2012 and 3.2% in 2016.^{7,8} Nigeria is considered a high HIV burden country in Sub-Saharan Africa.⁹ The 2018 HIV/AIDS survey showed that the North Central zone was the second highest region with a prevalence of 2.0%.⁷ It has been stated that Kogi State (in the north central zone) now has a prevalence of 0.9%⁷. Most people in the state are not aware of their HIV status and consequently not on antiretroviral treatment. This suggests that the rate of transmission might remain alarming with attendant ocular complications.

Young adults constitute a good percentage of all new HIV infections, which means they spend many years with the virus with the possibility of developing ophthalmic complication(s) that may progress to blindness

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and in the long run affect their quality of life.¹⁰⁻

¹¹ About 90% of people living with HIV/AIDS are in the developing countries, yet fewer studies on ocular problems in these group of patients exist compared to developed countries.¹² The nature and prevalence of ophthalmic problems in HIV-infected patients in developing countries, differ from those in developed countries.¹²⁻¹³ Emina et al. reported that cytomegalovirus retinitis is more dominant in the developed countries whereas in the developing countries herpes zoster ophthalmicus is more prevalent.¹³

Kogi State is an intersection state with a continuous influx of people of different HIV statuses to the state. Some of the surrounding states are high HIV burden states like Benue (with a prevalence of 5.3%)⁷ and their proximity could complicate the already high prevalence. Furthermore, Lokoja strategically connects the southern part to the northern region of Nigeria, and most of the long-distance commercial drivers make Lokoja a rest stop station. This encourages the booming of commercial sexual worker activities, which also increases the risk of HIV infection in the state. Obioma et al.¹⁴ report that such an attitude goes with attendant high-risk sexual behavior, which may result in HIV infection. Despite a high number of people with HIV/AIDS in the state (prevalence of 0.9%),⁷ no data on ocular manifestation exist. Hence, this study is done with the aim of creating awareness, planning, and mobilizing state resources in reducing the surge and ocular manifestations of HIV/AIDS in Lokoja.

MATERIALS AND METHODS

The study was a hospital-based cross-sectional study of adult HIV/AIDS patients on HAART attending the President's Emergency Plan for AIDS Relief (PEFAR) at Federal Medical Centre (FMC) Lokoja. The study was done over 4 months (May–August 2020).

The sample size of the study was calculated using the formula for population <

10,000 (the total population of HIV patients that have been registered in the hospital is 4001 over the years (1996 to 2020).

$$nf = \frac{n}{(1 + \frac{n}{N})}$$

Were

nf = final sample size

n

= sample size when the population is more than 10,000 (using a prevalence of [26.3%] ^15)

$$n = 298 \text{ (Calculated from } n = \frac{z^2 Pq}{d^2} \text{)}$$

were

n = minimum sample size

z = confidence interval of survey results; 95% ($Z=1.96$)

P = proportion of target population with the characteristics.

$q = 1 - P$

d = precision; 5%)

$$\text{Therefore, } nf = \frac{298}{(1 + \frac{298}{4001})}$$

$$nf = \frac{298}{1.07}$$

$$nf = 278.5$$

Approximately 279

Correcting for an attrition rate of 10% = 27.9, the minimum sample size would be 307. However, 310 participants were recruited.

A systematic random sampling technique was used to select eligible participants at the clinic. A random number was selected from the patient's attendant clinic record, and subsequently every fourth patient was selected until a sample size of 310 was completed. Inclusion criteria were consenting adults (≥ 18 years), HIV/AIDS patients that attended the PERFAR-supported ART clinic, and HIV/AIDS adult patients who were on HAART at the ART clinic of the hospital. Excluded from the study were adult HIV/AIDS patients with impaired mental states, and adult HIV/AIDS patients on HAART who declined consent.

All the participants' eyes were systematically examined using Snellen VA

chart (distance and near), ophthalmoscope, slit lamp and dilated fundoscopy when necessary. The socio-demographic characteristics (age, sex, religion, occupation, residence, marital status, education and family structure), duration of the disease, year on using HAART, and all the information got from eye examination were filled into pretest structured questionnaires.

Some participants noticed to have posterior segment lesions had their fundus photography done with a smartphone. Furthermore, automated perimetry was also done for all patients with glaucomatous disc damage and suspected neuro-ophthalmic pathology. All the participants had blood samples analyzed for CD4+ cell count and viral load.

The data was analyzed using Statistical Package for Social Sciences (SPSS) version 21, and the data was presented in Tables and figures. A p-value of < 0.05 was used to define statistical significance.

Approval for the study was obtained from the Ethics and Research Committee of FMC Lokoja. Furthermore, informed/thumbprint and written consent was obtained from all the participants.

RESULT

A total of 310 participants participated in the study, comprising 61 (19.7%) males and 249 (80.3%) females. The male to female ratio was 1:3.1. The age range was 21 years to 72 years, with most participants (68%) within the age range of 30 to 50 years, and the mean age was 41.29 ± 9.82 . Most of the participants (78.1%) had at least secondary education. Most of them were traders (46.8%) and civil servants (25.5%). The majority were married (84.55), practiced Christianity (66.1%) and resided in urban areas (70.3%). The participants' socio-demographic data is shown in table 1.

Table 1: Socio-demographic characteristics of participants N =310

Characteristics	Frequency (N)	Percentage (%)
Age (Years)		
21-30	51	16.5
31-40	108	34.8
41-50	103	33.2
51-60	39	12.6
61 and above	9	2.9
Mean age=		
41.29±9.82		
Gender		
Male	61	19.7
Female	249	80.3
Marital status		
Single	19	6.1
Married	262	84.5
Divorced	3	1
Separated	1	0.3
Widowed	25	8.1
Family structure		
Monogamous	231	74.5
Polygamous	79	25.5
Education		
Primary	50	16.1
Secondary	132	42.6
Tertiary	110	35.5
No formal education	18	5.8
Occupation		
Civil servant	79	25.5
Trading	145	46.8
Artisan	18	5.8
Unemployed	47	15.2
Driver	13	4.2
Paramilitary/Military	7	2.3
Others	1	0.3
Religion		
Christianity	205	66.1
Islam	103	33.2
Residence		
Rural	92	29.7
Urban	218	70.3

*others; retired civil servant

Duration of HIV/AIDS disease in years among participants

About 34.8% of the participants had the disease for less than 5 years. About 65.2% have had the disease for more than 5 years, as shown in the above figure 4 below.

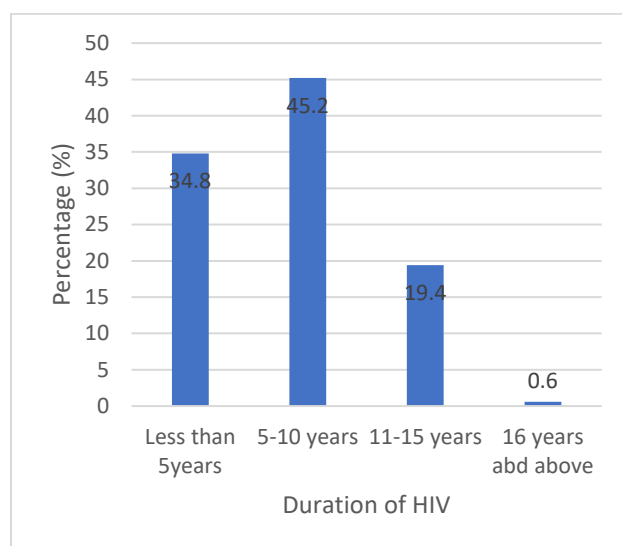


Figure 1: Duration of HIV/AIDS disease in years among participants Mean =6.61SD =4.04;95% CI: Lower =6.171; upper=7.045

Duration on HAART in months among participants

Figure 2 shows the duration of HAART in months among participants.

All the participants were on HAART, and 50.3% of them had been on HAART for more than 60 months (5 years).

Prevalence and pattern of ocular manifestations of HIV among the participants

The prevalence of ocular manifestations of HIV/AIDS in this present study was 5.2% (16 of the 310 participants had ocular manifestations). The commonest ocular manifestation noted was anterior uveitis (1.3%), and the least was cytomegalovirus retinitis (0.3%). The posterior segment had most of the ocular manifestations.⁹ There was no orbital manifestation seen in the study. The

two participants with squamous cell carcinoma (SCC) had surgery (excision biopsies and 5FU application); the SCC diagnosis was confirmed by pathology report. Other participants with HIV ocular manifestations were treated accordingly. The pattern of ocular manifestations is shown in table 2.

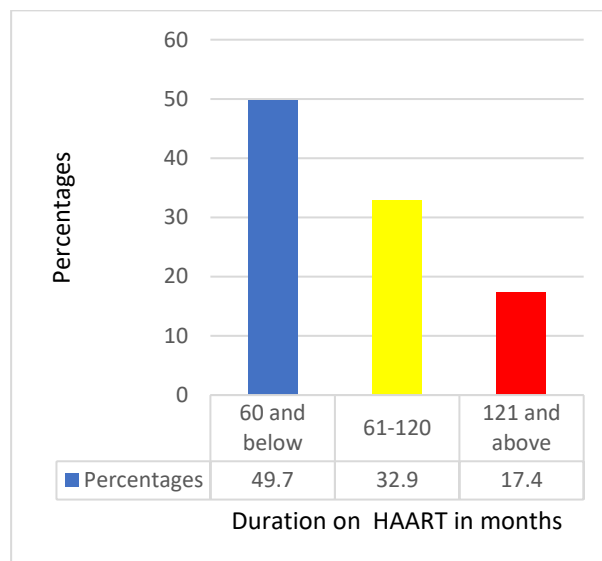


Figure 2: Duration on HAART in months among Participants; Mean = 74.593, SD= 46.31; 95% CI: Lower =69.407 upper =79.936

DISCUSSION

The prevalence of 5.2% ocular manifestations of HIV/AIDS in this study was low when compared with other studies.¹⁶⁻¹⁸ The possible explanation may be that all the participants recruited were on HAART, and most (50.3%) had been on HAART for more than 5 years. Additionally, most of the study participants (93.7%) were at HIV clinical stages I and II which were the stages in similar studies with low prevalence.^{2, 19} However, higher prevalence has been reported by some researchers^{16, 18} attributed to advanced clinical stages (WHO stages III and IV) and low CD4+ cell count of <200 cells/ μ l.¹⁶

A study in Brazil found no ocular manifestation of HIV/AIDS among the study participants. Small sample size (42 participants) and regular use of HAART were

Table 2: Prevalence and Pattern of ocular manifestations of HIV among the participants (N=16); and Viral load count level

<i>HIV related ocular manifestations</i>	<i>Frequency</i>	<i>%</i>	<i>Viral load count level</i>	
			<i>≤20</i>	<i>>20</i>
Adnexal Manifestation				2(100)
Active Herpes Zoster Ophthalmicus	2	0.65	0(0)	2(100)
Squamous Cell Carcinoma	2	0.65	0(0)	
Anterior Segment Manifestations				3(75)
Anterior Uveitis	4	1.3	1(2)	
Posterior Segment Manifestations				2(66.7)
Immune Recovery Uveitis	3	1	1(33.3)	2(100)
Toxoplasma Retinitis	2	0.65	0(0)	1(100)
Cytomegalovirus Retinitis	1	0.3	0(0)	
Neuro-ophthalmic Manifestation				0(0)
Optic atrophy	2	0.65	2(100)	
Total	16	5.2		

given as reasons.²⁰

In this study, the adnexa manifestations were noted to be 25% of all the ocular manifestations, similar to 26% reported by Listo et al.²¹ which found adnexa manifestations to be 26%. The same similarity with anterior segment manifestations was also reported by Listo et al (26.5%) and the current study (25%). Contrary to the findings of this study, Bekele et al.²² reported that adnexa manifestations made up 12.8% and anterior segment manifestations constituted 12.3%. The difference may be because of differences in the sample size and methodology of our study.

Posterior segment manifestations were the most commonly reported ocular manifestations of HIV/AIDS by Kim et al.²³, similar to the finding of this study (37.5%). Posterior segment lesions have been linked with CD4+ cell levels less than 200 cells/μl and increased viral load or virologic failure.²³ However, this study showed that most of the participants had a CD4 count > 200 cells/μl; due to HAART use. Studies by Bekele²² and Kim²⁴ also established the trajectory that

HAART's compliance is associated with increased CD4+ cell.

Conventionally, HAART is usually commenced when CD4+ counts are low. There is a possibility too that before patients commence HAART, posterior segment manifestations may have begun. Thirdly, newly formed CD4+ cells may lack functional maturity to provide the much-needed immune cover to patients. These are the likely factors that explained the higher prevalence of posterior segment manifestations with seemingly high CD4+ cell levels in our study. Low neuro-ophthalmic manifestations of HIV/AIDS appear to be common across studies.^{18, 25} This was the finding too in the current study. The orbit is rarely involved in HIV disease, as noted by some researchers.²⁶⁻²⁷ Herpes Zoster Ophthalmicus (HZO), which constituted 12.5% of the ocular lesions in our study, was noted to be a common adnexa manifestation. Adepoju et al.²⁸ reported that 50% of HIV-positive patients had HZO, and Nwosu et al.²⁹ reported that 48% of the HIV-positive patients studied had HZO. Umeh et

al.³⁰ in their study, observed that 50% of patients presenting with HZO were HIV positive and therefore suggested that HZO may be a pointer to underlying HIV infection. In our study anterior uveitis was the commonest ocular manifestation similar to the finding by Gogri et al.³¹ However, posterior segment manifestations were the commonest findings by Listo et al.²¹ The difference may be because of differences in the sample size and the race of our study population.

In our study, anterior segment ocular manifestations occurred at a higher CD4+ cell level compared with posterior segment manifestations. Few studies^{23,31} have reported this surprising phenomenon in which despite tangible CD4+, anterior segment diseases still soar. This study showed that 18.8% of all ocular manifestations were immune recovery uveitis (IRU) and 6.3% were cytomegalovirus retinitis (CMVR). While previous studies noted posterior segment findings such as toxoplasma retinitis,^{5,17,19} and posterior uveitis^{18,25,32} to be more common. Sudharshan et al.¹⁶ observed that 17.4% of those with ocular problems had IRU and 36.2% had CMVR. Gogri et al.³¹ observed that only 5% of those with HROM had IRU and 12.5% had CMVR. Bekele et al.²² did not observe any case of CMVR in their study but reported an IRU prevalence of 0.3%. IRU, though mainly associated with CMVR, has been documented to occur in association with other opportunistic infections.¹⁶

This study noted optic atrophy constituted 12.5% of all ocular manifestations and was the only neuro-ophthalmic manifestation seen, whereas Sudharshan et al.¹⁶ and Omolase et al.³³ reported optic atrophy as the commonest neuro-ophthalmic manifestation. Gaze palsies^{6,19} pupillary abnormalities³⁴, papilloedema,^{18,35} retrobulbar optic neuritis³⁶ and ptosis³⁷ were noted to be other neuro-ophthalmic findings by some researchers. However, these were not observed in the present study.

In conclusion, the prevalence of ocular manifestation of HIV/AIDS among HIV-positive patients attending our hospital is low. Anterior uveitis and immune recovery uveitis were the commonest ocular manifestations of HIV/AIDS seen in our patients. However, it is prudent for ophthalmologists to examine all HIV/AIDS patients in order to prevent visual impairment or blindness from HIV-related ocular complications.

Conflict of interest

The authors declare nil conflict of interest.

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